Discrete Distributions

Variable	Key Properties	$\mathrm{P}[X=x]$	$\mathrm{E}[X]$	$\mathrm{Var}[X]$	mgf
Uniform on $\{1,2,\ldots,n\}$		$\frac{1}{n}$ $1 \le x \le n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$	
Bernoulli	0 or 1 $1-p,\ x=0$	$p,\ x=1$	p	$p(1-p)$	pe^t+q
Binomial	Sum of Bernoullis number of successes in n trials	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$(pe^t+q)^n$
Geometric on $\{1,2,\ldots\}$	Number of trials until first success	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-qe^t}$
Geometric on $\{0,1,\ldots\}$	Number of failures before first success	$p(1-p)^x$	$\frac{1}{p}-1$	$\frac{1-p}{p^2}$	$\frac{p}{1-qe^t}$
Negative Binomial	Sum of Geometric on $0,\ldots$ Number of failure until r-th succss	$\binom{x+r-1}{r-1}p^r(1-p)^x$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{p}{1-qe^t}\right)^r$
Poisson	Sum of Poissons is Poisson	$e^{-\lambda}\cdot\frac{\lambda^n}{n!}$	λ	λ	$e^{\lambda(e^t-1)}$

Bernoulli Variance Shortcut: If X is a random variable that can only take on 2 values, with $\mathrm{P}[X=a]=p$ and $\mathrm{P}[X=b]=1-p$, then $\mathrm{Var}[X]=(b-a)^2p(1-p)$

Continuous Distributions

One trick to remembering the Gamma distribution is to recall that if α is an integer, then a Gamma(α, θ) distribution is the sum of α exponentials with mean θ.

Densities are only given for the region on which they are positive, and are 0 otherwise.

Distribution	range	density	cdf	EX	Var$X$	MGF
Uniform on $(0,1)$	$0 < x < 1$	1	x for $0 < x < 1$	$\frac{1}{2}$	$\frac{1}{12}$	$\frac{e^t - 1}{t}$
Uniform on (a,b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
Exponential with “rate” λ and mean $\theta = \frac{1}{\lambda}$	$0 < x < \infty$	$\lambda e^{-\lambda x} = \frac{1}{\theta} e^{-x/\theta}$	$1 - e^{-\lambda x} = 1 - e^{-x/\theta}$	$\theta = \frac{1}{\lambda}$	$\theta^2 = \frac{1}{\lambda^2}$	$\frac{1}{1-\theta t} = \frac{\lambda}{\lambda - t}$
Gamma	$0 < x < \infty$	$\frac{x^{\alpha-1}}{\theta^\alpha(\alpha-1)!} e^{-x/\theta}$	see below	$\alpha\theta$	$\alpha\theta^2$	$\left(\frac{1}{1-\theta t}\right)^\alpha$
Standard Normal	$-\infty < x < \infty$	$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$	$\Phi(x)$	0	1	$e^{t^2/2}$
Normal(μ, σ^2)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	σ^2	$e^{t\mu + (t^2\sigma^2)/2}$

For $\alpha = 1$, a Gamma is an exponential with mean θ and so has CDF $P[X \le t] = 1 - e^{-t/\theta}$

For $\alpha = 2$, a Gamma has cdf $P[X \le t] = 1 - e^{-t/\theta} - \left(\frac{t}{\theta}\right) e^{-t/\theta}$

For $\alpha = 3$, a Gamma has cdf $P[X \le t] = 1 - e^{-t/\theta} - \left(\frac{t}{\theta}\right) e^{-t/\theta} - \left(\frac{(t/\theta)^2}{2}\right) e^{-t/\theta}$

Note that each time we increase α by 1, we subtract another term from the cdf that looks that the probability distribution of a Poisson. In general, if α is an integer, the cdf of a Gamma(α, θ) is given by $P[X \le t] = P[X < t] = 1 - P[N \le \alpha - 1] = P[N \ge \alpha]$ where N is a Poisson random variable with mean $\lambda = t/\theta$.